Anuj Singh

SOFTWARE ENGINEER - BACKEND SYSTEMS, APIS & AI INTEGRATION

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Bangalore Urban, Karnataka, India

PROFESSIONAL SUMMARY

Software engineer with 2+ years building backend services and AI systems end to end. Ships strictly typed APIs and microservices in **Python, Node.js/TypeScript and Go**, designs PostgreSQL and Redis schemas, and tunes queries until latency drops - cut average API response time **40%** on YojanaSetu through PostgreSQL query optimization and frontend code-splitting. Integrates directly with AI model providers (OpenAI, Hugging Face, Pinecone) and external business systems over REST APIs, including ERP connectors and Stripe billing. Runs **Docker and Kubernetes on AWS** with CI/CD, and writes Pytest coverage before shipping. Merged pull requests in **canonical/cloud-init**, avoid-ai-writing, rtk, rea and code-graph-rag.

WORK EXPERIENCE

The Boring Education

Jan 2024 - Present

Data Scientist

Bangalore Urban, Karnataka, India

- Reduced model deployment time by **40%** with MLOps practices (CI/CD, monitoring, model versioning) on Docker, Kubernetes and AWS
- Built and maintained production data pipelines in Python (Pandas, Scikit-learn) feeding predictive models that triggered automated intervention workflows, increasing student retention **25%**
- Published 3 internal analytics frameworks adopted across product, marketing and curriculum teams, standardizing experimentation and KPI tracking
- Mentored 2 junior data scientists and established code review and documentation standards, improving team velocity and model reproducibility

Thiranex - Skill Development & Future Tech

Jul 2026 - Aug 2026

Data Science Intern

- Developed end-to-end machine learning workflows including data collection, cleaning, training, cross-validation and evaluation
- Built and deployed predictive analytics models in Python (Pandas, Scikit-learn, NumPy) across structured and unstructured datasets
- Improved model accuracy **18%** through systematic feature engineering and hyperparameter tuning
- Built interactive dashboards (Matplotlib, Seaborn) that turned raw datasets into findings non-technical stakeholders could act on

PROJECTS

YojanaSetu - Full-Stack AI Platform & API Backend

FastAPI backend serving a RAG eligibility engine (LangChain + Pinecone vector search + Hugging Face embeddings) at **94% accuracy** across 500+ documents, with a multilingual voice agent in 12 regional languages. Cut average API response time **40%** through PostgreSQL query tuning, indexing and Next.js code-splitting.

Python, FastAPI, PostgreSQL, Redis, Next.js, TypeScript, LangChain, Pinecone, AWS

AutoInvoice OCR - Micro-SaaS with External System Integration

Document pipeline extracting structured JSON from scanned invoices at **96% field-level accuracy** on non-standard layouts. Shipped end-to-end Stripe billing, ERP API connectors and a confidence-scoring review UI with human-in-the-loop validation.

Python, AWS Textract, Tesseract OCR, Next.js, Supabase, Stripe

Agent Parliament - Multi-Agent System on LLM Provider APIs

Architected a 5-agent debate framework with distinct persona prompts and a consensus synthesis engine using weighted voting, emitting structured decision documents with confidence scores. Integrates directly against OpenAI's model APIs with retry and streaming handling.

Next.js, React, TypeScript, OpenAI API, LangChain

DevRel GitHub Agent - Autonomous Service on GitHub Webhooks

Autonomous Probot GitHub App that classifies incoming issues by semantic analysis, auto-labels them and drafts PRs for trivial fixes - designed to resolve **~25%** of routine issues without human intervention.

TypeScript, Node.js, Probot, OpenAI API, GitHub Actions

Causal Inference Toolkit - Packaged Python Library

Production-ready Python library unifying DoWhy identification/refutation with EconML heterogeneous effect estimation, plus quasi-experimental engines (Synthetic Control with SLSQP, 2x2 and TWFE Difference-in-Differences). Ships a zero-code Streamlit dashboard, an automated report generator and Pytest coverage.

Python, DoWhy, EconML, Streamlit, Pytest

Open Source Contributions Merged upstream

Merged fixes in **canonical/cloud-init** (apt foreign-architecture package checks, VMware IMC type annotations, plus 7 further type-annotation PRs open), **conorbronsdon/avoid-ai-writing** (4 detector fixes), **rtk-ai/rtk**, **morluto/rea** and **vitali87/code-graph-rag**. Open PRs in review in hcengineering/platform, msitarzewski/agency-agents, megadose/holehe, shiyu-coder/Kronos and Zackriya-Solutions/meetily.

Python, TypeScript

EDUCATION

Bachelor of Engineering (BE), Computer Science & Engineering Chandigarh University 2026

CERTIFICATIONS

Project Planning and Machine Learning	University of Colorado Boulder (Coursera) | coursera.org/verify/N0O5KQNJM8K3
React Native	Meta (Coursera) | coursera.org/verify/MQUI4TAY8HR8

SKILLS

Languages: Python, TypeScript, JavaScript, Go, SQL

Backend & APIs: Node.js, FastAPI, Flask, REST APIs, API Design, Microservices, Webhooks

Databases & Caching: PostgreSQL, Redis, SQL, Supabase, Pinecone, Vector Databases, Query Optimization

Cloud & Infrastructure: Amazon Web Services (AWS), Docker, Kubernetes, CI/CD, GitHub Actions, Linux, Vercel

AI Integration: OpenAI API, LangChain, LangGraph, Hugging Face, RAG, LLM Orchestration, NLP

Data Engineering: ETL Pipelines, Data Pipelines, Pandas, NumPy, Scikit-learn, PyTorch, AWS Textract

Testing & Practice: Pytest, Cross-Validation, Model Evaluation, Experiment Tracking, Git, GitHub, Code Review

Frontend: React, Next.js, Tailwind CSS, HTML5, CSS3